

Notes: Veldkamp (AER, 2006)

Media Frenzies in Markets for Financial Information

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1 Big picture

- **Question.** Why do some financial markets experience sudden price run-ups and unusually high cross-market price dispersion at exactly the times when media coverage is intense?
- **Setting.** The paper studies emerging equity markets and proposes a model in which investors buy information in a competitive information market before choosing asset demand.
- **Core challenge.** Standard asset-demand complementarities are hard to sustain because assets are in fixed supply and market-clearing prices offset simple demand spillovers. Veldkamp instead looks for complementarities in *information demand*.
- **Main theoretical result.** Because information is nonrival and has high fixed cost but near-zero marginal cost, its price falls when more people demand it. This creates *information market complementarity*: investors want to buy the same information others buy because it becomes cheap.
- **Main asset-pricing result.** When information becomes abundant, it reduces the conditional variance of asset payoffs and raises prices. Switches between a no-information equilibrium and a high-information equilibrium generate *media frenzies*.
- **Main multiple-market result.** With several markets and a limit on how much information investors can absorb, information becomes concentrated in a subset of markets. Those markets receive more news, have higher prices, and generate *herding* in the form of excess cross-market price dispersion.
- **Main empirical contribution.** The paper builds a new panel of *Financial Times* news stories for 23 emerging markets and shows three empirical patterns consistent with the theory:
 - asset payoff volatility generates news;
 - more news is associated with higher asset prices;
 - more news is associated with greater cross-market price dispersion.
- **Economic takeaway.** Herds and frenzies can arise not because investors ignore information, but because they purchase too much of the same information when it becomes cheap. The source of the complementarity is the information market, not the asset market.

2 Data and measurement (key objects)

2.1 Observed data

- The empirical analysis uses a panel of **23 emerging equity markets** from **1989 to 2002**.
- The sample contains **weekly observations** on:
 - an asset price index,
 - a total return index,
 - and the number of news stories about each country.
- Total sample size is about **12,217 market-week observations**.

- Price and return data come from the **S&P 500 / IFCI Emerging Markets Database**.
- News is measured using **Financial Times** stories that mention the country name or adjective form in the title or lead paragraph.
- The paper uses only one news source to avoid double-counting and because the *Financial Times* provides a large fraction of relevant stories.
- From a random sample of 100 selected *Financial Times* stories, **97** were judged to contain information about the country's market or the value of its assets.

2.2 Asset, payoff, and news measures

- The **price index** P_{mt} measures country m 's market capitalization.
- The **total return index** R_{mt} measures market capitalization plus cash dividends.
- Because the model's risky asset is effectively two-period lived, the total return index is treated as the empirical analogue of the model's asset payoff.
- Dividend data are constructed as a **six-month backward-looking moving average**, denoted div_{mt} , because raw country-level dividends are cyclical and contain zeros.
- The key information variable is

$news_{mt}$ = number of Financial Times stories about country m in week t .

- Cross-market price dispersion is measured each week as the cross-sectional standard deviation of country price indices:

$$\left(\frac{1}{M} \sum_{m=1}^M (P_{mt} - \bar{P}_t)^2 \right)^{1/2},$$

where $\bar{P}_t$ is the average price across markets.

2.3 Key descriptive facts

- The 23 markets enter the sample at different dates, depending on when investable price indices become available.
- Across all countries:
 - the mean number of weekly news stories is **15.2**,
 - the mean asset price index is **255.0**,
 - and the standard deviation of the price index is **237.2**.
- News intensity varies a lot across countries:
 - China and Russia have more than **50** stories per week on average,
 - while Morocco averages only about **2.1**.
- Price levels also vary strongly across markets; for example, Argentina's average price index is much higher than that of China or Egypt.

- A useful scale fact from the price regressions is that one standard deviation of news is about **8.2 stories**. The paper interprets this as economically meaningful in the price-level regressions below.

2.4 Unobservables and timing

- The key latent object in the model is the persistent component of next period's asset payoff, θ_{t+1} .
- Investors differ only in whether they buy information. Informed investors observe θ_{t+1} before trading; uninformed investors do not.
- The risky asset also has an unobserved supply shock x_t , which prevents the price from fully revealing θ_{t+1} to uninformed investors.
- Timing within each period:
 1. investors decide whether to produce information and at what price;
 2. investors decide whether to buy information;
 3. informed and uninformed investors demand the risky asset;
 4. payoffs and shocks are realized.

3 Models and empirical specifications (all equations + interpretation)

3.1 One-market baseline setup

- Time is discrete. In each period, a large finite number of two-period-lived agents are born.
- Agents have CARA preferences over next-period wealth:

$$U(W_{t+1}) = E_t \left[-e^{-aW_{t+1}} \right].$$

- There are two assets:
 - one risky asset with payoff u_{t+1} ,
 - one riskless asset with payoff $r > 1$.
- The risky payoff has a persistent component and an idiosyncratic component:

$$u_{t+1} = \theta_{t+1} + \varepsilon_{t+1}.$$

- The persistent component follows a multiplicative AR(1)-type process:

$$\theta_{t+1} = (1 - \rho)\mu + \rho\theta_t(1 + \eta_{t+1}),$$

with

$$\varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2), \quad \eta_{t+1} \sim N(0, \sigma_\eta).$$

- **Interpretation.** Expected payoffs and payoff volatility move together because shocks are multiplicative. High-value assets also have more value at risk, so information about them is especially valuable.

3.2 Information structure and information markets

- In period t , an informed agent observes θ_{t+1} before trading.
- Uninformed agents observe only public history and the equilibrium price.
- The per-capita supply of the risky asset is

$$x_t \sim N(E[x], \sigma_x^2),$$

and is never directly observed. This hidden supply shock keeps prices from perfectly revealing fundamentals.

- Information is produced with a **fixed cost** χ and a **zero marginal cost of replication**.
- A representative information producer solves

$$\max_{d_{it}, c_{it}} d_{it} (c_{it} I(c_{it}, c_{-it}) - \chi),$$

where $d_{it} \in \{0, 1\}$ indicates whether agent i discovers information and $I(\cdot)$ is per-capita demand for information.

- **Interpretation.** News is expensive to discover but cheap to copy. This is the central source of complementarity.

3.3 Asset demand and equilibrium price

- Given an information set $\mathcal{F}_t^i$ and equilibrium price P_t , agent i demands

$$D(\mathcal{F}_t^i, P_t) = \frac{E[u_{t+1} \mid \mathcal{F}_t^i, P_t] - rP_t}{a \text{Var}[u_{t+1} \mid \mathcal{F}_t^i, P_t]}.$$

- If a fraction λ_t of investors buy information, market clearing is

$$\lambda_t D(\mathcal{F}_t^I, P_t) + (1 - \lambda_t) D(\mathcal{F}_t^U, P_t) = x_t.$$

- **Interpretation.** Information affects the price only through risk. By reducing the conditional variance of payoffs, it raises demand for the risky asset and therefore raises the equilibrium price.

3.4 Net benefit of information and information complementarity

- The expected net benefit of purchasing information is

$$B(\lambda) = \left[\frac{\text{Var}(u_{t+1} \mid P_t)}{\text{Var}(u_{t+1} \mid \theta_{t+1})} \right]^{1/2} - e^{ac(\lambda_t)}.$$

- The first term is the informational benefit: how much the price leaves unresolved relative to direct observation of θ_{t+1} .
- The second term is the utility cost of paying for information.

- **Proposition 1.** The equilibrium price of information $c(\lambda)$ is decreasing in the quantity of information purchased.
- **Proposition 2.** The expected net benefit $B(\lambda; \theta_t)$ is increasing in the fraction of informed agents over some range.
- **Interpretation.** This is the paper's key complementarity:
 - when more investors buy information, the fixed cost is spread over more buyers, so price falls;
 - but as information becomes abundant, the asset price itself becomes more revealing, so the marginal value of an extra signal falls.
- Because of these two opposing forces, the net-benefit curve is nonmonotone. This generates multiple information-demand equilibria and abrupt shifts between them.

3.5 Cutoff for information provision and the frenzy mechanism

- **Proposition 3.** There exists a cutoff θ^* such that:

$$\lambda_t^e > 0 \quad \text{for all } \theta_t \geq \theta^*, \quad \lambda_t^e = 0 \quad \text{for all } \theta_t \leq \theta^*.$$

- **Corollary 1.** The fraction of informed investors is nondecreasing in payoff variance.
- **Economic intuition.** High expected payoffs imply more value at risk. Since information has increasing returns in asset value, a sufficiently strong market can support a positive-information equilibrium, while a weak market cannot.
- **Proposition 4.** The expected price of the risky asset is strictly increasing in information provision:

$$E[P_t \mid \theta_t, \lambda_t] \text{ rises with } \lambda_t.$$

- **Corollary 2.** When $\theta_t \geq \theta^*$, the asset price is higher than it would be in a world without information markets.
- **Interpretation.** A *media frenzy* is not just a response to stronger fundamentals. It is the extra price increase created when fundamentals become strong enough to push the market into a high-information equilibrium.

3.6 Numerical example and excess volatility

- The calibration is chosen so that:
 - no-information and positive-information equilibria can both exist,
 - the stochastic process for θ_t matches the persistence of emerging-market prices,
 - the scale of shocks roughly matches price fluctuations in the data.
- The parameter values used in the paper's benchmark simulation are:

$$\chi = 0.012, \quad E[x] = 6, \quad \sigma_x^2 = 1.6, \quad \sigma_\eta = 0.09, \quad \sigma_\varepsilon = 0.1, \quad \rho = 0.99, \quad a = 2.5.$$

- Simulations show

$$\frac{\text{Var}(P^{info})}{\text{Var}(P^{noinfo})} = 1.5.$$

- Relative to a fixed-information-price model, allowing information prices to vary endogenously raises price volatility by about **40 percent**:

$$\frac{\text{Var}(P^{info})}{\text{Var}(P^{G-S})} = 1.4.$$

- **Interpretation.** The model produces long stretches of calm followed by abrupt price surges. This is the sense in which it captures *frenzies*: not steady overreaction, but regime-switching in information provision.

3.7 Multiple markets and herding

- The paper then extends the environment to M markets, each with an independent payoff process.
- To make markets interact, each investor can observe at most k signals. This creates a trade-off in learning about one market versus another.
- **Proposition 5 (Herding).** Suppose $\tilde{m}$ markets have identical histories and parameters, and let $\lambda^* = \arg \max_{\lambda} B(\lambda)$. If

$$\lambda^* > \frac{k}{\tilde{m}},$$

then equilibrium prices are not identical across markets; markets with more information provision have higher asset prices.

- **Mechanism.** When many markets are attractive, the information-capacity constraint binds. More news about one market crowds out information about others. This creates:
 - information trade-offs,
 - asset-price trade-offs,
 - and excess cross-sectional price dispersion.
- **Interpretation.** Herding here does not mean people blindly copy trades. It means they buy the same information, which then makes them buy the same assets.

3.8 Linking the model’s information demand to news stories

- The theoretical model uses λ , the fraction of informed investors.
- The data use the number of *Financial Times* stories.
- To bridge the two, the paper introduces a monopolistic-competition model of information varieties. When demand for information rises, more stories are produced.
- In the “varieties of the same information” version, the equilibrium number of stories satisfies

$$q \propto \lambda_t^{1/2}.$$

- In the “varieties of different information” version, each story is a noisy signal. Stories remain complements because the fixed cost of discovery falls with the scale of information production.
- **Interpretation.** The number of FT stories is not literally the price of information. Instead, it is a proxy for how easy and cheap it is to access information about a market from the low-cost media layer.

3.9 Empirical specifications

The main reduced-form specifications are:

1. **Risk $\rightarrow$ News**

$$news_{mt} = \tilde{c}_m + \tilde{\beta} (\Delta \log R_{mt})^2 + \tilde{\varepsilon}_{mt}.$$

2. **News $\rightarrow$ Price**

$$P_{mt} = c_m + \beta news_{mt} + \varepsilon_{mt}.$$

3. **News $\rightarrow$ Cross-market price dispersion**

$$\left(\frac{1}{M} \sum_{m=1}^M (P_{mt} - \bar{P}_t)^2 \right)^{1/2} = c + \beta \frac{1}{M} \sum_{m=1}^M news_{mt} + \varepsilon_t.$$

- Country fixed effects are included in the price-level regressions because country price indices are not directly comparable in levels.
- Payoff volatility is instrumented using dividend measures in the risk-to-news regressions.
- The paper also uses price-dividend ratios, Granger-causality tests, and simultaneous-equation estimation to strengthen the case that news affects prices rather than merely comoves with them.

4 Identification and instruments (what makes the empirical evidence credible)

4.1 What the theory predicts and what can be tested

- The model makes three sharp empirical predictions:
 1. higher payoff risk should generate more news;
 2. more news should be associated with higher asset prices;
 3. more news should be associated with greater cross-market price dispersion.
- These predictions are useful because alternative herd models, especially information-cascade models, would typically predict that herds emerge when information is *scarce*, not abundant.

4.2 What identifies the effect of risk on news

- The cleanest empirical relation in the paper is the effect of payoff volatility on news.
- The identifying idea is that **dividends** are good instruments for payoff volatility:
 - dividend changes are largely predetermined outcomes of earlier decisions;
 - they predict payoff volatility;
 - and newspaper publication does not itself affect dividends.
- The paper instruments payoff volatility with:

$$div_{mt}, \quad (\Delta div_{mt})^2, \quad \text{and lags of payoff volatility.}$$

- A Granger-causality test with 25 lags also rejects the null that lagged payoff volatility does not predict news.
- **Interpretation.** News is treated as endogenous, not exogenous. Asset-market conditions help generate media coverage.

4.3 What supports the effect of news on prices

- There is no clean instrument for news itself, because asset-payoff-relevant news is, by construction, closely tied to asset prices.
- The paper therefore uses several complementary pieces of evidence:
 - price-level regressions,
 - price-dividend ratio regressions,
 - Granger-causality tests,
 - simultaneous-equation estimation,
 - and robustness checks controlling for payoff volatility.
- The logic for using the price-dividend ratio is especially important: high P/D should not itself mechanically create more news through the model's risk channel, so a positive news- P/D relation is more informative than a simple news-price relation.

4.4 Why the dispersion result is especially persuasive

- The price-dispersion regression is closest to a direct test of the *herding* mechanism.
- If information were merely more abundant everywhere, prices of all assets would tend to rise together.
- The key prediction of the model is stronger: when information becomes concentrated in only a subset of markets, cross-market price dispersion should rise.
- This is harder to rationalize with fixed information costs or standard information-cascade stories.

4.5 Limits of the empirical strategy

- The paper is not a full structural estimation of the dynamic model.
- Some of the central empirical evidence is contemporaneous, so strict causal interpretation is limited.
- In particular, the paper is transparent that there is no valid instrument for the effect of news on prices.
- Still, the combination of:
 - instrumented volatility $\rightarrow$ news,
 - positive news $\rightarrow$ price and P/D ,
 - positive news $\rightarrow$ price dispersion,
 - and robustness to controls,

makes the overall pattern highly consistent with the model.

4.6 Relation to the literature

- **Grossman–Stiglitz.** The paper keeps the Grossman–Stiglitz logic that information matters because prices are not fully revealing, but it endogenizes the price of information.
- **Information cascades / herd models.** It contrasts with Banerjee (1992), Bikhchandani, Hirshleifer, and Welch (1992), and related cascade models. In those models, herd behavior is associated with too little information revelation; here, high information supply itself can create frenzies.
- **Liquidity.** The paper also connects to liquidity-based interpretations: abundant, cheap public information can lower the risk of trading against better-informed agents and thus behave like endogenous liquidity provision.
- **Macro-finance asymmetry.** The model also provides a microfoundation for boom-bust asymmetry driven by endogenous information flows.

5 Tables and figures

5.1 Figure 1: expected net benefit of information

Read:

- Figure 1 is the conceptual heart of the paper.
- The curve for the net benefit of information is initially rising and then falling.
- The rising part comes from price complementarity: more buyers make information cheaper.
- The falling part comes from strategic substitutability: once information is abundant, the price already reveals a lot, so an extra signal is less valuable.

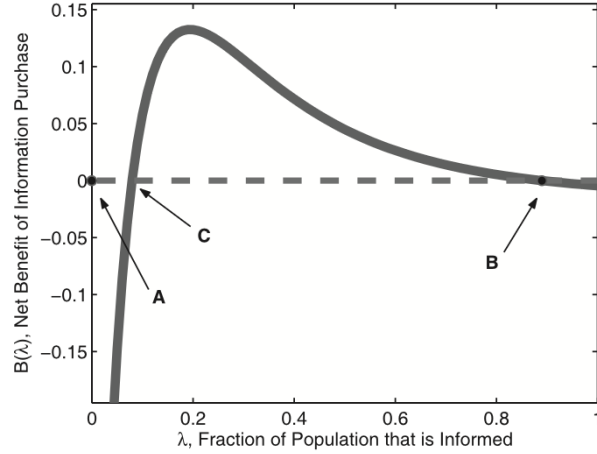


FIGURE 1. EXPECTED NET BENEFIT OF INFORMATION, AS THE AGGREGATE DEMAND FOR INFORMATION VARIES

Notes: When information supply is low, the benefit of information is increasing because its price is falling. When information is abundant, the benefit to information is declining because the asset price reveals more of the information. The parameter values used to generate the graph are: $\sigma_\theta = 0.1$, $\sigma_\varepsilon = 0.1$, $\sigma_x^2 = 10$, $a = 0.7$, $\chi = 1.8$, $n = 50$.

- The figure shows multiple candidate equilibria for information demand, but only the high-information point is also an equilibrium in information supply.
- This nonmonotonicity is exactly what makes discrete jumps in information provision possible.

5.2 Figure 2 and Table 1: simulated frenzies and excess volatility

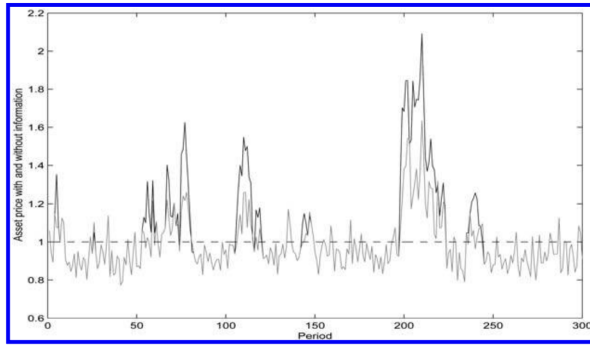


FIGURE 2. SIMULATED ASSET PRICES IN THE NO-INFORMATION MODEL (LOWER LINE) AND INFORMATION MODEL (UPPER LINE)

Note: Excess price volatility: $\text{Var}(P_{\text{info}})/\text{Var}(P_{\text{no info}}) = 1.5$.

TABLE 1—PRICE VOLATILITY IN A BENCHMARK NO-INFORMATION MODEL, IN THIS MODEL, AND IN EMERGING MARKETS

	No-information model	Information model	Emerging market data
Price standard deviation	0.29	0.39	0.43
$\text{Var}_{(Price)}$	1.0	1.8	1.3
$\text{Var}_{(Price(Low\ News))}$			

Notes: Low news is defined as a level of λ , or count of news stories that falls below its median. Simulated price changes are log changes: 5,000 simulations. Emerging market data is log price indices for 23 emerging markets described in Section VI. Moments computed on a country-by-country basis, starting at country start date (see Table 2) and ending in June 2002, and then averaged across countries.

Read:

- Figure 2 compares simulated asset prices in the information model and in a no-information benchmark.
- The key visual message is that prices can sit close to the benchmark for long periods and then

suddenly jump upward when the market moves into a high-information equilibrium.

- Table 1 quantifies the volatility effect:
 - price standard deviation is **0.29** in the no-information model,
 - **0.39** in the information model,
 - and **0.43** in the emerging-market data.
- The ratio of unconditional variance to low-news variance is:
 - **1.0** in the no-information model,
 - **1.8** in the information model,
 - **1.3** in the data.
- The point is not that the model matches every moment exactly, but that it can plausibly generate a large enough volatility effect to explain observed frenzies.

5.3 Figure 3 and Figure 4: information crowd-out and cross-market herding

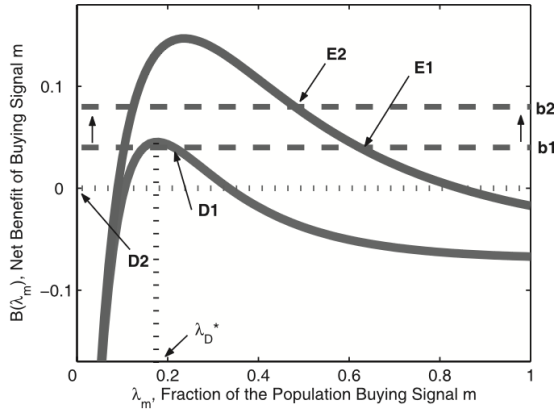


FIGURE 3. INFORMATION CROWD-OUT

Notes: Higher demand for information in a third market (not in graph) raises the net benefit cutoff level from $b1$ to $b2$. Information demand in market E falls from $E1$ to $E2$. Information demand in market D falls to zero.

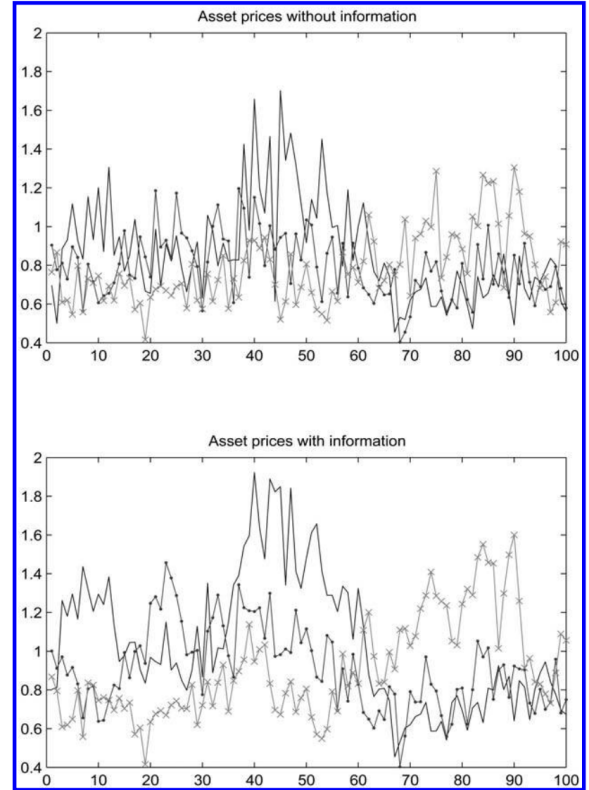


FIGURE 4. PRICE OF THE RISKY ASSET IN THREE MARKETS WHERE INFORMATION TRADE-OFFS ARE PRESENT

Notes: In the bottom panel, the information market drives a wedge between the markets covered in the media and the noncovered markets. The result is cross-sectional variance in price that is 1.5 times higher in the market with information. This simulation uses the same parameter values as the simulation in Section III. The number of signals an agent can purchase each period is one and the total number of markets is three.

Read:

- Figure 3 illustrates the information crowd-out mechanism.
- When one market becomes more attractive from an information-demand perspective, the equilibrium cutoff net benefit rises and information demand in other markets falls.
- Some markets may be pushed all the way to a no-information equilibrium.
- Figure 4 then shows the price implication: in otherwise identical markets, the market that receives more information trades at a higher price.
- In the simulated three-market example, cross-sectional price variance is **1.5 times** higher with information trade-offs than without.
- This is the model's cleanest notion of *herding*: investors stampede from one market to another through the news market.

5.4 Table 2: descriptive statistics for the dataset

Read:

- Table 2 shows large heterogeneity in media coverage and price levels across the 23 emerging markets.
- The average number of weekly news stories across all countries is **15.2**, but there is substantial cross-country variation.
- Average price indices also vary a lot, with an all-country mean of **255.0**.
- The differing start dates remind the reader that this is an unbalanced panel: markets entered the investable universe at different times.

TABLE 2—DESCRIPTIVE STATISTICS FOR DATASET

Country	Number of news stories				Asset price index		
	mean	min	max	st dev	Start date	mean	st dev
Argentina	12.5	0	71	8.5	January 1989	695.8	321.4
Brazil	19.1	0	118	10.8	January 1989	278.5	143.9
Chile	6.7	0	32	4.4	January 1989	500.0	203.8
China	50.2	14	114	14.5	September 1993	61.3	25.7
Czech Republic	9.6	1	47	5.7	April 1996	55.0	10.1
Egypt	6.4	0	47	5.7	February 1997	62.9	25.3
Greece	11.8	0	38	5.9	January 1989	378.4	216.8
Hungary	9.5	0	29	4.7	January 1994	225.1	67.0
India	25.8	2	85	9.8	November 1992	90.7	20.9
Indonesia	13.7	0	87	10.5	September 1990	67.8	39.2
Israel	22.3	6	78	9.2	August 1997	138.9	26.1
Malaysia	12.0	0	50	7.5	January 1989	174.7	82.3
Mexico	19.0	0	71	10.8	January 1989	587.6	219.4
Morocco	2.1	0	13	2.0	February 1997	135.4	24.5
Peru	4.5	0	30	3.4	September 1993	169.8	39.2
Philippines	8.6	0	44	5.8	January 1989	159.7	84.1
Poland	13.6	2	49	6.2	January 1994	601.3	145.2
Russia	51.9	17	193	20.5	February 1997	87.4	41.9
South Africa	29.3	5	71	10.9	January 1994	188.2	39.5
South Korea	21.2	0	82	11.4	January 1992	87.7	31.9
Taiwan	10.7	1	33	6.3	January 1991	126.0	31.6
Thailand	10.9	0	54	8.3	January 1989	184.3	123.7
Turkey	11.1	0	48	6.1	August 1989	208.5	116.3
All countries	15.2	0	193	13.9		255.0	237.2

Notes: Asset price data start on date listed and end in June 2002. The variable div_{mt} begins six months after the start date. News data descriptive statistics summarize the sample period from the asset price start date to June 2002.

TABLE 3—ASSET MARKET RISK INCREASES NEWS PROVISION

Dependent variable	Independent variable	Regression coefficient	With dividend instruments
$news_{mt}$	log payoff volatility	58.40	196.81
	$\Delta \log(R_{mt})^2$	(12.08)	(11.44)
$news_{mt}$	payoff volatility	1.93	10.73
	$(R_{mt} - R_{mt-1})^2 \times 10^{-4}$	(0.23)	(1.45)
$news_{mt}$	payoff	3.26	5.07
	$R_{mt} \times 10^{-3}$	(0.27)	(0.42)
mean ($news_{mt}$) in market m	log payoff standard deviation	2.45	
$(1/T) \sum_{t=1}^T news_{mt}$	$((1/T) \sum_{t=1}^T (\log(R_{mt}) - \log(\bar{R}_m))^2)^{1/2}$	(0.39)	

Notes: Regression coefficients in rows 1–3 result from estimating (6) with seemingly unrelated regressions. The coefficient estimates with dividend instruments result from 3SLS using the following instruments: div_{mt} , $(\Delta div_{mt})^2$, and four lags of the independent variable. Row 4 contains an OLS estimate of equation (7). SUR estimates use all available data, starting at the start date for each country listed in Table 2 and ending in June 2002: 12,194 observations. The instrumented regressions use div_{mt} , which begins six months after the country start date: 11,596 observations. All tables show standard errors in parentheses.

5.5 Table 3: asset market risk increases news provision

Read:

- This is the first core empirical table.
- News is strongly positively related to measures of payoff risk.

- With log payoff volatility, the SUR coefficient is **58.40**; with dividend instruments, it rises to **196.81**.
- Using payoff volatility in levels also gives positive coefficients: **1.93** and **10.73** with instruments.
- The cross-country regression also shows that markets with greater payoff standard deviation receive more news, with coefficient **2.45**.
- This table supports the claim that news is endogenous and generated by risky market conditions.

5.6 Table 4 and Table 5: news and asset price levels

TABLE 4—NEWS AND ASSET PRICE LEVEL

Dependent variable	SUR coefficient on $news_{mt}$
Asset price	0.28 (0.03)
Price-dividend ratio	0.18 (0.02)

Notes: Equation (8) is estimated using seemingly unrelated regressions, to allow for contemporaneous correlation across countries and heteroskedasticity. Data cover from country's start date (Table 2) to June 2002: 12,194 observations.

TABLE 5—THE EFFECTS OF NEWS ON PRICE AND OF PAYOFF VOLATILITY ON NEWS, JOINTLY ESTIMATED

Equation I	Coefficient on $news_{mt}$
Dependent variable	1.35 (0.03)
P_{mt}	
Equation II	Coefficient on $(\Delta \log(R_{mt}))^2$
Dependent variable:	211.98 (21.63)
$news_{mt}$	

Notes: Estimated using 3SLS. The equations estimated are (8) and (6). The following instruments are used for $(\Delta \log(R_{mt}))^2$: div_{mt} , $(\Delta div_{mt})^2$, and $(\Delta \log(R_{m(t-1)}))^2$. Data cover from six months after country's start date (Table 2) to June 2002: 11,596 observations for each equation.

Read:

- Table 4 shows that more news is associated with higher prices and higher price-dividend ratios.
- The coefficient of news on the price index is **0.28**; on the price-dividend ratio it is **0.18**.
- The paper interprets this economically as follows: a one-standard-deviation increase in news (8.2 stories) coincides with about a **\$2.30** increase in the country price index, roughly **1%** of the average price level.
- Table 5 jointly estimates the news and price equations using 3SLS and allows cross-equation error correlation.
- The coefficient of news on price rises to **1.35**, while the coefficient of payoff volatility on news is **211.98**.
- The paper's Thailand example is useful for interpretation:
 - average FT stories rise from **3.8** per week (1989–1992) to **19.8** (1996–1997),
 - the average price index rises from **164** to **243**,
 - and the paper argues that roughly **28%** of that price increase could be attributed to news if the model forces are operative.

5.7 Table 6 and Figure 5: news and cross-market price dispersion

Read:

TABLE 6—TEST OF THE HERDING HYPOTHESIS:
PRICE DISPERSION INCREASES WHEN EMERGING
MARKET NEWS IS ABUNDANT

Dependent variable	Constant	OLS coefficient on news
Price dispersion $((1/M) \sum_{m=1}^M (P_{mt} - \bar{P}_t)^2)^{1/2}$	84.35 (7.78)	8.41 (0.49)

Notes: Coefficients result from OLS estimation of equation (9); 704 weekly observations, January 1989–June 2002.

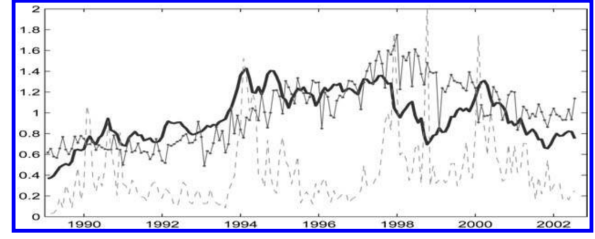


FIGURE 5. AVERAGE PRICE INDEX (SOLID LINE), NUMBER OF NEWS STORIES (DOTS), AND
PAYOFF VOLATILITY (DASHED LINE)

Notes: Monthly data. All series are normalized by their respective country means before they are averaged across all emerging markets.

- Table 6 is the direct empirical test of the herding hypothesis.
- The coefficient of news on cross-sectional price dispersion is **8.41** with standard error **0.49**.
- Even restricting to the shorter period where all markets have price data, the coefficient remains positive and significant (**2.82**).
- Figure 5 plots average price index, average number of FT stories, and payoff volatility over time.
- The figure and the appendix unit-root tests address a concern that the results may simply reflect common trends in prices and news. The paper argues that neither the price series nor the news series display the type of strong trend that would mechanically generate the correlations.

6 Short summary

- **Method.** The paper builds a dynamic asset-pricing model with endogenous information production and decreasing information prices due to high fixed costs and free entry.
- **Mechanism.** Information is nonrival, so when more investors demand the same information, it becomes cheaper. This generates information complementarity, abrupt jumps in information provision, and media frenzies.
- **Multiple markets.** With limits on how much information investors can absorb, more attention to one market crowds out attention to others, causing cross-market herding and excess price dispersion.
- **Evidence.** In 23 emerging markets, payoff volatility predicts news, news predicts higher prices, and news predicts greater price dispersion across markets.
- **Main insight.** The source of frenzies and herds can lie in the information market rather than in asset-demand spillovers or investor irrationality.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that a competitive market for information can itself generate frenzies and herds.

- It establishes that the nonrival nature of information is enough to create complementarities in information acquisition.
- These complementarities produce regime switches between low-information/low-price and high-information/high-price outcomes.

7.2 Economic intuition

- Investors do not rush into the same asset because they directly like imitating one another.
- Instead, they rush toward the same *information*. Once that information becomes cheap, they all become better informed about the same market and therefore all want to hold the same asset.
- Cheap information reduces conditional risk and raises prices, especially in markets whose fundamentals are already strong enough to support positive information demand.
- With multiple markets and limited attention, information concentration in one market mechanically means information scarcity in others, which produces cross-market price wedges.

7.3 Takeaway

This paper is important because it offers an alternative to standard herd and bubble stories. Instead of relying on irrationality, short horizons, or non-market-clearing assumptions, it shows how ordinary competitive forces in a market for information can generate large swings in prices and media coverage. The deeper lesson is that when information has high fixed costs and low marginal costs, financial markets can become unstable not because information is absent, but because it becomes too concentrated in the same places at the same time.